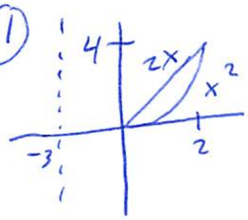
1452 Final Solutions

Spring 2014

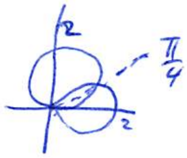
① 

a) washers $\int_0^4 \pi(\sqrt{y}+3)^2 - \pi(\frac{1}{2}y+3)^2 \, dy$

b) shells $\int_0^2 2\pi(x+3)(2x-x^2)\,dx$

c) $M_x = \int_0^2 \frac{1}{2}(2x+x^2)(2x-x^2)\,dx$

2) $\int_0^3 \sqrt{1+(-2x+2)^2}\,dx$

3) 

$$\int_0^{\frac{\pi}{4}} \frac{1}{2}(2\sin\theta)^2 d\theta + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{2}(2\cos\theta)^2 d\theta$$

④ a) $\int x \ln x \, dx \quad = \frac{1}{2}x^2\ln x - \int \frac{1}{2}x\,dx = \frac{1}{2}x^2 \ln x - \frac{1}{4}x^2 + C$

$u = \ln x \qquad dv = x\,dx$

$du = \frac{1}{x}dx \qquad v = \frac{1}{2}x^2$

b) $\int \frac{\csc^2(3x)}{\cot(3x)}dx \quad = \frac{-1}{3}\int \frac{1}{u}du = \frac{-1}{3}\ln|\cot(3x)| + C$

$u = \cot(3x)$

$du = -3\csc^2(3x)\,dx$

c) $\int \frac{1}{\sqrt{x^2-4}}dx \quad = \int \frac{1}{\cancel{2}\cancel{\tan\theta}} \cancel{2}\sec\theta \cancel{\tan\theta}\,d\theta \quad = \int \sec\theta\,d\theta$

$x = 2\sec\theta$

$dx = 2\sec\theta\tan\theta\,d\theta$

$\ln|\sec\theta + \tan\theta| + C$

$\ln\left|\frac{x}{2} + \frac{\sqrt{x^2-4}}{2}\right| + C$

d) $\int \frac{2x-5}{(x+3)(x+4)}dx = \int \frac{A}{x+3} + \frac{B}{x+4}\,dx \quad = \int \frac{-11}{x+3} + \frac{13}{x+4}\,dx$

$A(x+4) + B(x+3) = 2x-5 \qquad = -11\ln|x+3| + 13\ln|x+4| + C$

$\underline{x=-4} \qquad 0 - B = -13$

$\underline{x=-3} \qquad A + 0 = -11$

(5) $\int_0^1 \frac{3}{x+2}\,dx$ is <u>not</u> an improper integral. Although $\frac{3}{x+2}$ has a vertical asymptote at $x=-2$, that is not between the limits of integration.

6) To find $\lim_{k\to\infty} k^{1/k}$, take log. $\lim_{k\to\infty} \frac{1}{k}\ln k = \lim_{k\to\infty} \frac{\ln k}{k} = 0$.

Thus $\lim_{k\to\infty} k^{1/k} = e^0 = 1$.

7) (a) Since $\sum c_k (x-5)^k$ converges at $x=8$, its radius of convergence must be at least 3. It must converge on $(2,8)$, and in particular, must converge at $x=4$.

(b) Since $\lim \frac{a_k}{\frac{1}{k^2}} = 3$ and $\sum \frac{1}{k^2}$, then $\sum a_k$ conv. by Limit Comp. Test.

8) (a) $\sum \frac{k^k}{3^k}$ diverges b/c $\lim_{k\to\infty} \frac{k^k}{3^k} = \infty$ Divergence Test

<u>or</u> $\lim_{k\to\infty} \sqrt[k]{\frac{k^k}{3^k}} = \infty$ Root Test

(b) $\sum \frac{(\ln k)^2}{k}$ diverges b/c $\frac{(\ln(k))^2}{k} > \frac{\ln k}{k} > \frac{1}{k}$ Direct Comp. Test

<u>or</u> $\int_1^\infty u^2\,du = \infty$ Integral Test

(c) $\sum \frac{(-1)^k}{2k}$ converges (conditionally) b/c $\lim_{k\to\infty} \frac{1}{2k} = 0$ A.S.T.

but $\sum \frac{1}{2k}$ diverges

(d) $\sum \frac{k}{k^3+5}$ converges b/c $\frac{k}{k^3+5} < \frac{1}{k^2}$ Direct Comp. Test

<u>or</u> $\lim_{k\to\infty} \frac{\frac{k}{k^3+5}}{\frac{1}{k^2}} = 1$ Limit Comp. Test

9) note $\sum_{k=0}^{\infty} \frac{2}{k!}(x-3)^k = 2e^{x-3}$ and $R=\infty$

or $\lim_{k\to\infty} \frac{2|x-3|^{k+1}}{(k+1)!} \frac{k!}{2|x-3|^k} = \lim_{k\to\infty} \frac{|x-3|}{k+1} = 0$ for all x.

Thus $R=\infty$ and interval of convergence is $(-\infty, \infty)$

10) $x^3\cos(5x) = x^3 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!}(5x)^{2k} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} 5^{2k} x^{2k+3}$

11) $\vec{u} = \langle 1, 0, -3\rangle$ $\quad \vec{v} = \langle 0, -2, 1\rangle$

a) $\vec{u} - 3\vec{v} = \langle 1, 6, -6\rangle$

b) $\vec{u}\cdot\vec{v} = -3 = \sqrt{10}\sqrt{5}\cos\theta$ $\quad \theta = \cos^{-1}\left(\frac{-3}{\sqrt{10}\sqrt{5}}\right)$

c) $\frac{\vec{u}\times\vec{v}}{\|\vec{u}\times\vec{v}\|}$ will be $\perp$ to $\vec{u}$ and $\vec{v}$

$$\vec{u}\times\vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -3 \\ 0 & -2 & 1 \end{vmatrix} = \langle -6, -1, -2\rangle$$

$$\|\vec{u}\times\vec{v}\| = \sqrt{36+1+4} = \sqrt{41}$$

So $\left\langle \frac{-6}{\sqrt{41}}, \frac{-1}{\sqrt{41}}, \frac{-2}{\sqrt{41}} \right\rangle$ or $\left\langle \frac{6}{\sqrt{41}}, \frac{1}{\sqrt{41}}, \frac{2}{\sqrt{41}} \right\rangle$ will work.